

Hardware Upgrade Plan

Capacity forecast & buy list · Data measured Aug 13, 2026 · Prices checked Aug 13, 2026

THE ONE-PARAGRAPH VERSION

Output grew **~6.5× June → July** and is tracking **~4× again in August**. The DGX box (which is actually an **ASUS Ascent GX10**, confirmed from its board ID) is out of runway on **two fronts at once**: the boot drive has **17 GB free (99% full)** against 15–35 GB/day of render churn — that's **less than a week** — and memory is already over-subscribed (the full workload wants 111 GB of 119 GB, which is exactly what caused the Aug 9 lockup). Cloud overflow (Modal) is compounding toward **\$550–700 this month**. You don't have a future cap-out date. **You capped out this week**. The good news: **~\$1,055** fixes the emergencies, and **~\$3,100** more roughly doubles render capacity.

+554%

Video output June → July (110 → 719 files). July → Aug tracking +321% more.

17 GB

Free disk on the DGX boot drive (99% full) vs 15–35 GB/day of render churn. **< 1 week**.

0 slots

Real render headroom at full concurrency — workload wants 111 of 119 GB usable memory.

\$403

Modal cloud GPU spend month-to-date (day 13) — was \$128 on day 3. Pacing \$550–700.

THREE ANSWERS, UP FRONT

A One purchase, \$5,000

\$2,999–3,099

A second GB10 box (MSI EdgeXpert or ASUS Ascent GX10 — same machine as your DGX, 128 GB). Move the language-model server to it → frees 51 GB on box #1 → **~2× render capacity, doubles total RAM to 256 GB**, zero software porting. Pays for itself vs Modal in **~4.6–4.8 months**. Leaves ~\$1,900 of the budget for the emergencies on page 4.

B Multiple purchases

\$5,789 total

The four-step stack: ① **\$595** external 4 TB SSD (fixes the 99% disk *this week, zero downtime*) → ② **\$460** two pure-sine UPS units → ③ **\$3,099** second GB10 → ④ **\$1,635** UNAS Pro + 4×12 TB drives at 10 GbE. Full stack on page 4.

C The one you didn't think of

\$595 + \$0

External USB4 NVMe (4 TB WD SN850X + \$65 enclosure): ~2 GB/s on the Spark's USB-C, clears 400 GB off the boot drive with **no reimage, no downtime, no warranty risk**. Plus three **free** capacity moves (page 5) — the biggest one recovers up to 51 GB of memory without spending a dollar.

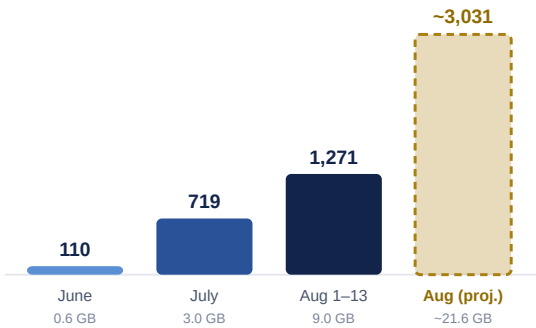
Verdict on the 5090 and the CPU rebuild: the RTX 5090 is real (2.75–4.7× your current render speed) but is scalped to \$3,700–4,600 right now — it's the **trigger-based later buy**, not the first dollar. The ASRock motherboard/CPU rebuild is a **skip**: with a GPU in that box, the i5-12600K is not the bottleneck, and a new CPU adds zero transcode capability (same Quick Sync engine). Details pages 3–5.

What the data shows

Every number below was measured on your systems on Aug 13, 2026 — file counts from disk, memory from the kernel, kills from the system logs, spend from the Modal billing collector.

Videos rendered per month

Files produced across all pipelines · August projected from days 1–13



Undercounts June/July: the 33-city listings pipeline deletes its videos after 3 days (~500–700 renders invisible to this count). The real curve is even steeper.

Modal cloud GPU spend — August, day by day

Month-to-date billing, last 10 daily readings · dashed = month-end projection



This is the bill that a GPU purchase can actually displace — the payback math on pages 3–4 is measured against it.

Where the 119 GB of usable memory goes at full workload

Why the Aug 9 lockup happened — and why today's calm is an illusion (2 of 3 render services are currently dead)



The 51 GB is reserved **even when idle**. 8 GB of slack for the OS + everything else = out-of-memory kills went 1/month (July) → 6 in the first 13 days of August.

The five gauges, side by side

RESOURCE	TODAY	STATUS
Boot drive	99% full · 17 GB free	CAPPED
Memory	111 of 119 GB wanted	CAPPED
CPU	load 20.3 / 20 cores	SATURATED
NAS Vol 1	88% · 1.4 TB free	MONTHS
Power backup	none · 47-min outage in June	EXPOSED

If growth merely continues at half pace

Even if the +321%/month curve cools to +100%/month:

- Disk** — already gone. Fails on churn within days, growth aside.
- Memory** — every render slot you add is consumed within ~1 month of growth. Owned capacity covers the base load; Modal stays the peak-shaver.
- Modal bill** — \$625 → ~\$1,250 (Oct) → ~\$2,500 (Nov) if nothing is bought. That's the number the buy list is racing.

Answer A — “One piece of hardware, \$5,000”

If exactly one box gets bought, this is the box.



Second GB10 machine — MSI EdgeXpert MS-C931 or ASUS Ascent GX10 (128 GB)

\$2,999–3,099

payback ~4.6–4.8 months vs Modal

The identical brain to your current DGX box, from an OEM instead of NVIDIA — same 128 GB unified memory, same software stack, ~\$1,600 cheaper than NVIDIA's own unit (which was raised to \$4,699 in February).

[ASUS Ascent GX10 — \\$3,099 · Amazon \(in stock\)](#) · [Newegg first-party](#) · [MSI EdgeXpert — \\$2,999 · MSI direct](#) · [Dell GB10 \(2 TB\) — \\$3,699](#)

Why it wins the single-slot: it attacks the actual bottleneck. Move the 51 GB language-model server to box #2 and box #1 gets **+2 render slots**; box #2's remaining ~68 GB adds **2–3 more**. Net: **+4–5 concurrent renders ≈ 2× local capacity**, total RAM doubles to 256 GB (the thing you said you wanted), bigger models become possible across both boxes, and the out-of-memory kill spiral ends. Zero porting — clone the working environment over. It's the lowest-risk upgrade that exists for this fleet.

Why not the RTX 5090 for the single \$5k slot? (It was close.)

	SECOND GB10 · ~\$3,099	RTX 5090 · \$3,695–4,600 STREET
Raw render speed	1× (same as today, ×2 boxes)	2.75–4.7× per job — genuinely faster
Fits the 51 GB LLM server	Yes — the whole point	No — 32 GB card can't host it
True cost to be useful	\$3,099 + \$15 cable	\$3,695–4,600 + \$699 RAM + possible \$250 PSU
Setup risk	None — clone existing stack	Linux reimage of the Windows box, PSU verification, connector-melt watch, scalped market climbing further
Payback vs Modal	~4.6–4.8 mo	~5.9–7.4 mo new · ~4.8 used

The 5090 is the right **second** move, on a trigger: buy used (~\$2,900–3,300) when the Modal bill crosses ~\$1,000/month, or on any dip. At today's scalper prices, the GB10 does more for the actual constraint per dollar. Full 5090 file: [live price tracker](#) · [Micro Center \(resists scalping — check the Overland Park store\)](#) · [used-market history](#).

Spend the leftover ~\$1,900 of the \$5k on page 4's steps ① and ② (\$595 storage + \$460 power protection = \$1,055) — they fix the two emergencies the GB10 doesn't touch, and you still come in ~\$850 under budget.

Set-up rules for box #2 (from your own Aug 9 post-mortem): it gets the same memory-guardrail treatment as box #1 on day one — the out-of-memory protector plus service memory caps — or it inherits the same lockup behavior. And both boxes plug into the battery backup from page 4 before it goes 24/7.

Answer B — The full stack, in strict order

Multiple purchases, ranked by what breaks first. Do them top to bottom; each step stands alone.

1

Fix the disk — this week

URGENT

\$595

WD_BLACK SN850X 4 TB + ACASIS USB4 enclosure, plugged into the DGX's USB-C. ~2 GB/s real speed — model loads go 3s → 7s, meaningless next to render times. Move the 398 GB model folder over; boot drive drops 99% → ~50%. **No reimage, no downtime, no warranty risk** (vs the internal swap: full wipe, 1–2 days of dead pipelines, warranty-voiding per ASUS).

pays back < 1 month in avoided Modal overflow

 [SN850X 4 TB — \\$539.99 \(Amazon / Best Buy / Newegg / Micro Center\)](#) · [ACASIS TBU405 enclosure — ~\\$65 · Amazon](#)

2

Power protection — two pure-sine UPS units

URGENT

\$460

2× \$220 · flagged “biggest single failure mode” since June

2× CyberPower CP1500PFCLCD (true sine wave — verify the listing says **1000W**; old revision is 900W). Unit 1: network shelf (gateway, router, switch, NAS) — rides out **25–40 min**, most of a repeat of June's 47-minute outage. Unit 2: compute shelf (DGX + towers) — 6–9 min, enough for an automatic clean shutdown before a dirty power cut kills a 99%-full drive mid-render. *Skip the UniFi-branded UPS — simulated sine wave.*

 [CP1500PFCLCD — \\$219.95 each · Amazon](#) · [Micro Center ~\\$240](#)

3

Double the compute — second GB10 (Answer A)

\$3,099

ASUS Ascent GX10 or MSI EdgeXpert, 128 GB. LLM server moves over; +4–5 render slots across the pair; 256 GB total RAM. Buy the cheap 1 TB version (this box's job needs ~60 GB of disk) + a ~\$15 QSFP cable for the box-to-box link.

payback ~4.6–4.8 mo vs Modal

 [ASUS GX10 — \\$3,099 · Amazon](#) · [MSI EdgeXpert — \\$2,999](#)

4

Bulk storage — UNAS Pro + 4× 12 TB, at 10 GbE

THIS QUARTER — prices rising

\$1,635

\$538 + 4×\$264 + \$60 · ≈ \$45 per usable TB

Ubiquiti UNAS Pro (7-bay rackmount, 10G) + **4× Seagate IronWolf 12 TB new** = 36 TB usable (RAID5), room to grow to 60 TB. Plus the ~\$60 10G kit (DAC for the NAS + 10GBASE-T module so the DGX actually links at 10G — without it everything runs at 2.5G). **Buy drives at order time:** the AI-datacenter drive shortage has prices up ~50% and rising; recertified drives are currently a trap (page 6). Supersedes the old ~\$200–370 Plex JBOD idea — point Plex at the UNAS instead. *2U rackmount — needs rack space or a shelf. If the 8-bay UNAS Pro 8 (\$799) is in stock at order time it's a worthy step-up; don't wait on it.*

 [UNAS Pro — \\$499 \(\\$538 w/ surcharge\) · ui.com](#) · [IronWolf 12 TB — \\$264 each · Newegg](#) · [UNAS Pro 8 — \\$799 \(if in stock\)](#) · [10G DAC + module — ~\\$60](#)

Full stack, all-in — ①+② buy survival · ③ buys growth · ④ buys a year of storage

\$5,789

Answer C — Free moves & the roadmap

Three \$0 moves worth real capacity (do before any box arrives)

MOVE	WHAT IT RECOVERS	COST
Shrink the language-model reservation	The LLM server pre-reserves 51 GB <i>even when idle</i> . A smaller/quantized model, a lower reservation setting, or on-demand loading recovers up to 51 GB — 2+ render slots — for free . This is the single biggest lever in the whole report; the second GB10 makes it permanent, but the config change works today.	\$0
Disk triage	~49 GB sits in old container images (prune), plus the 386 GB model folder likely holds checkpoints no live pipeline loads — auditing it buys weeks of runway before the SSD even arrives.	\$0
Fix the record-keeping	System logs only reach back ~2 days, so June's history is gone and this analysis rests on thinner data than it should. Raise log retention + start collecting GPU/memory metrics so the <i>next</i> hardware decision has a real trend line.	\$0

The full roadmap on one line each

WHEN	ACTION	COST
This week	External 4 TB SSD + move models off the boot drive · disk triage · LLM reservation trim	\$595
This week	2× pure-sine UPS, auto-shutdown wired to both shelves	\$460
This month	Second GB10 (ASUS/MSI) · LLM server moves to it · guardrails cloned on day one	\$3,099
This quarter	UNAS Pro + 4× IronWolf 12 TB + 10G links · Plex & archives migrate · old JBOD retires	\$1,635
On trigger	RTX 5090 (used dip or Modal > \$1k/mo) + 64 GB RAM into the ASRock tower, Linux reimage	~\$3,700
Ongoing	Modal stays as the elastic peak-shaver — owned boxes carry the base load, cloud absorbs spikes	shrinks

The trigger buy, spelled out

5

RTX 5090 into the ASRock tower — when the trigger fires

Trigger: Modal crosses ~\$1,000/month, or used 5090s dip under ~\$3,000. The tower and 1000W PSU are already owned; add the card + a 64 GB RAM kit, reimage to Linux, power-limit the card to 450W. 2.75–4.7× per-render speed, ~200 clips/day ceiling on that one box. Verify the PSU has a native 12V-2x6 cable first; if not, +\$250 for a 1200W ATX 3.1 unit.

Used-market tracker (\$2,868–3,590) · FE new — \$3,695 · Newegg · Corsair 64 GB DDR5-6000 — \$699

~\$3,700+

used card + RAM · payback ~4.8 mo at today's Modal, faster later

What NOT to buy — asked and answered

ITEM	WHY IT'S A SKIP
ASRock motherboard/CPU rebuild (i7-14700K \$381 / i9-14900K / new board)	With a GPU in that box, video generation is ~95% GPU-bound — the i5-12600K isn't the bottleneck. A new CPU adds zero transcode capability (identical Quick Sync engine on all three chips), the socket is dead-end, and the i9 class has a documented history of degrading under exactly your 24/7 duty cycle. The only real gap in that box is RAM, and only when the 5090 arrives.
NVIDIA-brand DGX Spark FE — \$4,699	Functionally identical to the \$2,999–3,099 OEM GB10 boxes. Price was raised \$700 in February; you'd be paying ~\$1,600 for the logo and a 4 TB drive you don't need.
96 GB RAM kit — \$1,173	RAM crisis pricing (up 300–400% since mid-2025). 64 GB at \$699 covers the 5090 build; 96 GB only if that box later takes on extra duty.
Recertified 12 TB drives — \$300	Currently inverted: costs <i>more</i> than new (\$264), shorter warranty, and out of stock anyway. The 2023-era “recert bargain” playbook is dead this cycle.
UniFi UPS Tower / 2U — \$159/\$279	Simulated sine wave + tiny batteries — incompatible with the DGX power brick and the 1000W PSU. Pure sine is a hard requirement.
Internal 2 TB drive swap for the DGX — \$160–805	Forces a full wipe + reimage + 1–2 days of dead pipelines, voids warranty coverage per ASUS, and the 2242-size drives are scalped. The external route gets more space for less risk.

Sleeper option, for the record: the ASRock tower itself (6 drive bays, 1000W PSU, already owned) + a ~\$90 10G network card + the same 4 drives ≈ **\$1,150 for the same 36 TB** — the best raw dollars-per-terabyte of any storage option. Rejected only because that box is earmarked as the future 5090 render node and the UNAS is zero-maintenance. If the 5090 plan ever dies, revisit this.

Fine print — where every number came from

Measured on your hardware, Aug 13 2026

- **Growth:** file census across all render output folders, bucketed by month (June 110 / July 719 / Aug 1–13 1,271 videos; 0.59 / 3.02 / 9.04 GB). August projected linearly from 13 days. Known undercount: the listings pipeline auto-deletes media after 3 days.
- **Memory:** kernel accounting — 87 of 119 GB in use with 16 GB swapped *while 2 of 3 render services were down*; LLM server reservation 51.4 GB (a fixed pre-allocation); each render slot ≈ 20–25 GB.
- **Out-of-memory kills:** kernel logs — 1 in July, 6 in Aug 1–13; plus 365 GPU allocation failures and 36 CUDA out-of-memory errors in a single 48-hour window. June logs rotated away (hence the \$0 record-keeping fix).
- **Disk:** boot drive 853 of 916 GB (99%); model folder 398 GB; container storage 49 GB. NAS: 11 TB volume at 88%, 2 TB volume at 74%.
- **Spend:** Modal billing collector daily readings, \$128 → \$403 month-to-date over ~10 days; 30-day rolling total \$549. The machine identifies as an ASUS Ascent GX10 (GB10 family) via its board ID.

Prices — checked Aug 13, 2026 (volatile markets flagged)

ITEM	PRICE	SOURCE / NOTES
ASUS Ascent GX10 128 GB	\$3,099	Amazon first-party · avoid a \$5,099 third-party listing on Newegg
MSI EdgeXpert MS-C931 128 GB	\$2,999	GB10 buyer's guide · MSI direct; Amazon carries a 4 TB version ~\$3,999
WD_BLACK SN850X 4 TB	\$539.99	Multi-retailer tracker · NAND shortage: was ~\$250 in 2024; unlikely to drop soon
ACASIS USB4 enclosure	~\$65	Amazon · this exact chipset family is the one confirmed fast on the Spark's ports
CyberPower CP1500PFCLCD x2	\$440	Amazon \$219.95 · confirm "1000W" revision; >\$240 is a bad listing, not the market
Ubiquiti UNAS Pro (7-bay, 10G)	\$499 + \$39 surcharge	ui.com · checkout adds a tariff surcharge; budget \$538
Seagate IronWolf 12 TB x4	\$1,056	Newegg \$264/drive · drive shortage, up ~50% since Sept; price is perishable — buy with the enclosure
RTX 5090 32 GB	\$3,695–4,600	Live tracker · MSRP \$1,999–2,399, heavily scalped, further rises predicted; used \$2,868–3,590
Corsair 64 GB DDR5-6000 kit	\$699	Corsair direct · RAM crisis: ~4× 2025 prices; sub-\$250 listings are stale or scams

Key judgment calls (and how to overturn them)

- **GB10 over 5090 first** — the binding constraint is memory, not speed. Overturned if the LLM server gets retired, or 5090s return to ~MSRP.
- **External SSD over internal swap** — zero downtime beats a cleaner install. Overturned if a planned reimage happens anyway.
- **UNAS over more UGREEN** — best \$/bay with 10G, ends the fragile-NFS single point. Overturned if rack space truly doesn't exist.
- **"Buy soon" on drives/SSD/RAM** — all three markets are in AI-driven shortages with rising forecasts into 2027; waiting has a real cost this cycle.